

Chapter 158 — Active Management Strategies for Localised Prostate Cancer — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Very low risk** → **active surveillance (AS) best option**; low risk → AS for most.
- **Favourable intermediate** = 1 intermediate factor + grade group (GG) 1–2 + <50% cores.
- **Staging imaging from unfavourable intermediate up**: bone scan + CT/MRI.
- **proPSMA**: PSMA PET sensitivity **85% vs 38%**; accuracy 92% vs 65%.
- **ProtecT 15 yr**: cancer death ~2–3% all arms; more metastasis on AS.
- **SPCG-4**: RP vs watchful waiting RR **0.56**, NNT **8**; benefit <65, intermediate.
- **PIVOT**: no benefit low risk; 19.5 yr benefit intermediate risk only.
- **Epstein**: PSAD <0.15, Gleason ≤6, <3 cores, ≤50% any core.
- **RP** → incontinence + ED; **EBRT** → bowel; **brachy** → worst acute LUTS.
- **Intermediate/high risk** → **discourage AS** unless limited longevity.

HOST EVALUATION

- **Longevity tools (NCCN)**: Social Security tables or WHO Life Tables.
- **Charlson Comorbidity Index (CCI)**: built for 1-yr inpatient mortality.
- **PCCI** (prostate-specific CCI): predicts 10-yr mortality.
- **CAN score** beats CCI and PCCI for 5-yr overall survival.
- **PCOS**: 10-yr cancer death 3–18% vs ≥33% other-cause with any comorbidity.
- **Operate if**: high risk of metastasis/death within 10 yr, despite competing risks.
- **Comorbidity (cardiac, pulmonary, haematological)** → may contraindicate surgery, not treatment.

BASELINE FUNCTION

- **Instruments**: IPSS (International Prostate Symptom Score); UCLA Prostate Cancer Index.
- **Obstructive LUTS / large gland** → surgery; brachytherapy → retention.
- **Brachytherapy difficult** >60 cm³ (pubic arch interference).
- **Neoadjuvant ADT** shrinks gland 25–40%; may be needed if >80 g before RT.
- **TURP before, not after, EBRT**; post-brachy TUR → incontinence.
- **Delay RT 6–8 wk after TURP** → ↓ incontinence, bladder neck contracture.
- **Small gland <30 cm³ + large TURP defect** → avoid brachy/cryo.
- **Limited TUR** (median lobe/bladder neck) safe before brachytherapy.
- **Retention + radiation** → switch catheter to intermittent catheterisation.
- **Severe compliance abnormality** → surgery (RT fibroses bladder).
- **Absolute contraindication to RT**: previous radical pelvic irradiation.
- **Relative (RT)**: IBD, bowel in field/hernia, bowel-trapping surgery, poor healing (diabetes).
- **Relative (surgery)**: chronic steroids.
- **Urologist's recommendation** most influences choice.

RISK ASSESSMENT

- **Pillars**: clinical stage, biopsy Gleason, pretreatment PSA.
- **AUA/ASTRO/SUO**: 4 groups; **NCCN** splits intermediate (fav/unfav) and high (high/very high).
- **Very low**: T1c, GG1, PSA <10, <3 cores, ≤50%/core, PSAD <0.15.

- **Low**: T1–T2a, GG1, PSA <10.
- **Intermediate factors**: T2b–T2c, GG2–3, PSA 10–20.
- **Unfavourable intermediate**: 2–3 factors, or GG3, or ≥50% cores.
- **High**: T3a, or GG4–5, or PSA >20.
- **Very high**: T3b–T4, primary pattern 5, or >4 cores GG4–5.
- **Gleason 1966; ISUP 2005** → patterns 1–2 dropped, all cribriform = pattern 4.
- **Grade groups 2013 (ISUP 2014)**: 3+3=1, 3+4=2, 4+3=3, 4+4=4, 9–10=5.
- **Epstein insignificant**: organ-confined, <0.5 mL; accuracy 79%.
- **Epstein caveat**: validated on systematic, not MRI-targeted, biopsy.
- **Grade migration** → historical vs contemporary Gleason 6 not comparable.

STAGING

- **Gold standard for bone**: technetium-99m MDP bone scan; low specificity.
- **Equivocal bone scan** → PET/CT or MRI.
- **CT** misses metastasis in nodes <10 mm; still endorsed for nodal staging.
- **MRI** only test accurate for EPE/SVI; high PPV, poor NPV for EPE.
- **SVI on MRI**: T2-hypointense mass contiguous with base tumour.
- **Choosing Wisely**: ASCO 2012, AUA 2013 — no imaging in low risk.
- **proPSMA**: 302 high-risk; specificity 98% vs 91%; management change 27%.
- **18F-DCFPyL**: specificity 97.9%, sensitivity 40.3%; PPV 86.7 vs 28.3%.
- **AUA 2022**: PSMA after negative conventional; **NCCN 2023**: instead of it.

NOMOGRAMS, MRI, GENOMICS

- **Partin tables** (1993): PSA+Gleason+stage → pathological stage; 72.4% accurate.
- **Kattan**: AUC 0.79 for relapse after RP.
- **UCSF-CAPRA**: adds age, % positive cores; 5-yr RFS 85% → 8%.
- **PI-RADS v2**: ADC predicts progression and recurrence.
- **Tissue assays gold standard** over liquid biomarkers.
- **NCCN genomics**: AS candidates, low/fav intermediate, life expectancy >10 yr.
- **AUA genomics**: not routine; not in small-volume low risk.

COMPARATIVE STUDIES

- **Best end points:** cancer-specific and overall mortality, not biochemical.
- **SPCG-4:** pre-PSA era, ~90% palpable, 23.2 yr follow-up.
- **SPCG-4:** absolute cancer-death reduction 11.0 points.
- **PIVOT:** 731 men; bone metastases 4.7% vs 10.6%.
- **PIVOT 10 yr:** all-cause benefit only PSA >10.
- **PIVOT 19.5 yr:** intermediate risk ARR 14.5%.
- **PIVOT 19.5 yr overall:** all-cause ARR 5.5%, cancer 4.0% — both NS.
- **PIVOT flaws:** underpowered; ~20% non-adherent; ~40% dead by 10 yr.
- **PCOS:** RP overall HR 0.60, disease-specific 0.35 — selection bias.
- **ProtecT:** 1643 randomised; RT arm with 3–6 mo ADT.
- **ProtecT 10 yr:** cancer mortality ~1% vs 15% predicted → underpowered.
- **ProtecT cohort:** 77% low risk; only 6% GG ≥3.
- **ProtecT 15 yr:** 2.7 / 2.2 / 2.9%; AS crossover 60%.
- **ProtecT <65:** surgery beats RT for cancer mortality.
- **Sooriakumaran:** RT vs RP sHR 1.76 non-metastatic.
- **CEASAR:** RP sexual deficit only in top baseline quartile.
- **CEASAR low risk:** EBRT sexual score 12 points better at 3 yr.
- **Chen (JAMA 2017):** RP incontinence meaningful to 24 mo; EBRT bowel at 3 mo.
- **PCOS function:** RP vs RT differences vanish by 15 yr.
- **Open vs robotic:** equal function at 12 wk (one RCT).

STEPWISE COUNSELLING

- **Order:** treat? → partial vs whole → surgery vs RT vs ablation → delivery.
- **AS** = active testing, deferral; **watchful waiting** = symptom-driven, ADT at progression.
- **Occult high grade** in low-grade on systematic biopsy: 25–45%.
- **Reclassification predictors:** PSAD >0.15, core density; not race or age.
- **Any Gleason pattern 4** → ↑ eventual metastasis; low-volume GG2 AS debated.
- **NICE:** MRI mandatory for all AS; MRI at 12–18 mo.

- **Confirmatory biopsy** within 6–12 mo (NCCN, ASCO).
- **ASCO:** biopsy every 3–5 yr; NCCN: biopsy ≤ annually, PSA ≥6-monthly intervals.
- **Toronto** (inclusive) 15-yr cancer death 5.7% vs **Hopkins** (Epstein) 0.6%.
- **Toronto treatment:** 25 / 36 / 45% at 5 / 10 / 15 yr.
- **Hopkins treatment:** 50 / 57% at 10 / 15 yr.
- **Toronto:** non-prostate death 10× prostate cancer death.
- **PRIAS:** PSAD ≤0.2; PSADT triggers biopsy.
- **Canary** only programme admitting Gleason ≤7.
- **AS deaths:** mostly cardiovascular.
- **Focal therapy:** not for high risk; best for limited-focality intermediate risk; trials.
- **Cryo** → ED almost always; **HIFU** retreatment 29–38%.
- **RT late risk:** rectal/bladder cancer at 15–20 yr.
- **RP counselling:** early incontinence and impotence are the norm.
- **RT + ADT** (intermediate/high risk): ADT drives early toxicity.

CLASSIC TRAPS

- **Trap:** 10-yr life expectancy decides treatment — **Actually:** weigh cancer aggressiveness too.
- **Trap:** GG3 is favourable intermediate — **Actually:** GG3 → unfavourable.
- **Trap:** Gleason 8 = GG5 — **Actually:** 8 = GG4; 9–10 = GG5.
- **Trap:** CT excludes nodal disease — **Actually:** blind to normal-sized nodes.
- **Trap:** PSMA PET highly sensitive for nodes — **Actually:** DCFPyL nodal sensitivity ~40%.
- **Trap:** ProtecT shows AS equal on everything — **Actually:** more metastasis on AS.
- **Trap:** EBRT worst for acute LUTS — **Actually:** brachytherapy worst.
- **Trap:** TURP after radiotherapy is safe — **Actually:** TURP before; after → incontinence.
- **Trap:** observational RP advantage is causal — **Actually:** selection bias (PCOS 3-yr divergence).
- **Trap:** AS = watchful waiting — **Actually:** AS curative intent; watchful waiting palliative.